

PRICE SENSITIVITY ALONE DOES NOT EXPLAIN CUSTOMER CHURN

BCG X DATA SCIENCE (FORAGE) | CUSTOMER CHURN, PRICE SENSITIVITY, AND RETENTION PRIORITIES

9.72%

baseline churn

overall churn rate in PowerCo dataset

0.715

model ROC-AUC

meaningful risk-ranking signal

8.45%

churn recall

only 24 of 284 churners detected

0.507

price p-value

off-peak price change not significant

Recommended decision: pilot targeted churn interventions and improve detection before operational deployment.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

PowerCo should treat churn as a multi-signal retention problem rather than a simple price-sensitivity issue

The data suggests price movement alone is not enough to explain churn, while customer relationship, value, usage, and pricing context together create a stronger risk picture.

1 Price sensitivity was weak as a standalone explanation

Jan–Dec off-peak price change had near-zero correlation with churn ($r \approx -0.006$; $p \approx 0.507$).

2 Predictive signal exists, but classification is too conservative

Random Forest ROC-AUC was 0.715, but recall was only 8.45%, catching 24 of 284 churners.

3 Model predictors point to broader customer economics and behavior

Margins, consumption intensity, meter rent, forecast gaps, and tenure-related features ranked above most price-change metrics.

4 Recommended actions should be targeted and testable

Start with price-shock monitoring, renewal/product-depth interventions, and a behavior-based early warning pilot.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

The initial hypothesis tested whether price changes, especially off-peak changes, were driving churn

We structured the analysis as a business hypothesis test before moving into model building.

1. Define hypothesis

Customers exposed to meaningful price increases should churn more often.

2. Engineer price signal

Calculate January-to-December off-peak price change at customer level.

3. Compare churn rates

Group customers by price-change bucket and compare churn rate.

4. Model churn

Use engineered features to classify churn and identify broader signals.

5. Translate to action

Recommend retention plays only where data shows actionable risk.

Why this matters for the business

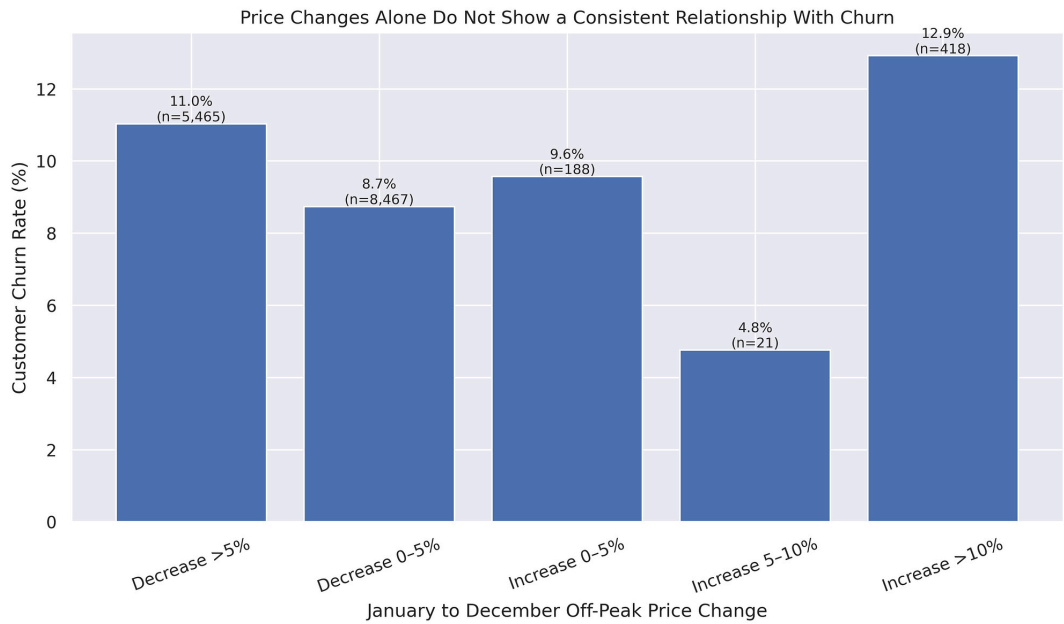
If price is the dominant driver, PowerCo might consider pricing interventions. If churn is multi-factorial, broad discounting risks sacrificing margin without reaching the highest-risk customers.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

FINDING 1

Off-peak price changes do not show a consistent churn pattern, weakening the broad price-sensitivity hypothesis

Customers with >10% off-peak price increases had elevated churn, but the full pattern is inconsistent and the direct correlation is not significant.



*The 5–10% increase bucket contains only **21 customers**, so its 4.8% churn rate should not be interpreted as a stable result. The overall conclusion remains valid.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

Fact base

Churn baseline	9.72%
Off-peak price-change r	-0.006
p-value	0.507
>10% increase group	12.9% churn
Largest decrease group	11.0% churn

FINDING 1 DETAIL

Current results do not support price change as a primary churn driver; more targeted forms of price sensitivity require further testing

Averages may hide the customer experience: sudden peaks, volatility, or prices above a personal threshold may matter more than annual averages.

What the data did not support

A consistent standalone link between annual off-peak price change and churn.

What remains plausible

Whether customers respond differently to short-term price shocks, volatility, unusually high prices, or pricing near renewal

What to test Next

Maximum monthly price changes, price volatility, customer-relative pricing, and timing between price changes and churn.

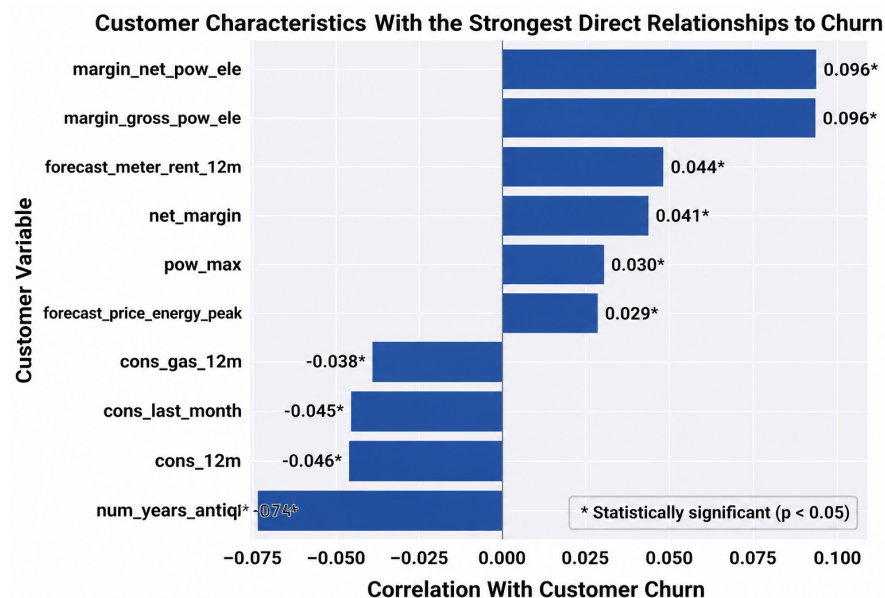
The current analysis does not rule out more targeted forms of price sensitivity. Annual price movement may obscure effects from short-term pricing events or customer-specific exposure, which should be tested separately before drawing a broader pricing conclusion.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

FINDING 2

Customer economics and tenure show larger but still weak, direct association with churn than the initial off-peak price-change feature

The most correlated customer variables are directionally small, but they provide a useful shortlist for deeper customer segmentation.



How to read this chart

- Positive $r \rightarrow$ higher values are associated with higher churn
- Negative $r \rightarrow$ higher values are associated with lower churn
- Magnitude matters more than the sign for strength
- All displayed correlations are small, so these are signals to investigate, not causal proof

Notable direct signals

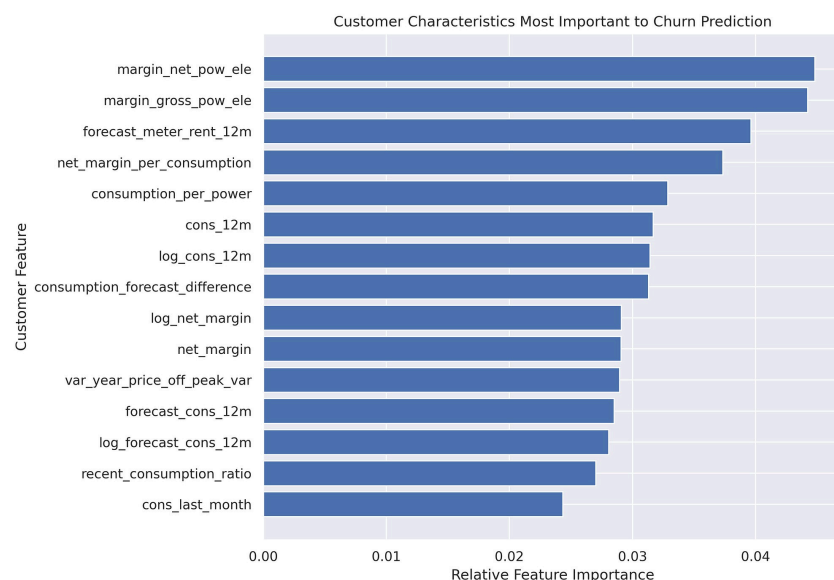
- Net and gross power margins trend with churn
- Tenure trends against churn
- Consumption measures trend against churn
- Peak forecast price shows modest positive relationship

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

FINDING 3

The model relies most on customer value, usage intensity, and forecast gaps, not just tariff changes

Feature importance broadens the churn story from price sensitivity to customer economics and behavior.



How to interpret

- Positive values are associated with higher churn
- Negative values are associated with lower churn
- All relationships remain small and should be treated as segmentation signals, not causal drivers

Signals worth deeper investigation

- Customer margin and profitability
- Tenure
- Consumption patterns
- Forecast pricing context

Source: PowerCo customer and pricing

datasets; Random Forest baseline on engineered features

MODEL PERFORMANCE

The current model is precise when it flags churn, but it misses most churners and is not deployment-ready

Accuracy appears high because churn is rare; recall reveals the operational limitation.

91.0%

Accuracy

overall correct predictions

88.9%

Precision

predicted churn is usually correct

8.45%

Recall

misses most churners

15.4%

F1 Score

low due to recall

0.715

ROC-AUC

useful risk ranking

	Predicted Stay	Predicted Churn
Actual Stay	2,635	3
Actual Churn	260	24

Confusion matrix highlights the business issue

The model correctly catches only 24 of 284 churners and misses 260. It makes very few false churn alerts, which explains the high precision, but this is not sufficient for proactive retention.

⊗ **Implication:** Use the model as a risk-ranking baseline for further validation, not as an automated churn-intervention engine

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

The model can support risk segmentation, but targeting value should be validated before operational use

PowerCo should convert the model output into prioritized retention queues and evaluate lift before scaling.

What the model demonstrates now

Rank customers by relative churn risk and identify a small high-confidence set.

Current Model Limitations

Reliably capture enough churners for a scaled retention program at the default threshold.

What to validate next

- Churn rate by risk decile
- Performance across customer segments
- Alternative probability thresholds
- Expected retention economics

Recommended operating approach

- Rank customers by predicted churn probability instead of relying only on the 0.50 classification cut-off.
- Pilot retention interventions in the highest-risk deciles and compare against holdout groups.
- Measure business lift: churn reduction, saved margin, incremental discount cost, and customer lifetime value.
- Tune the threshold based on retention budget and acceptable false-positive volume.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

Three targeted actions should be piloted if further data confirms the observed churn-risk themes.

1 Price-shock intervention

Hypothesis to validate: abnormal price spikes, volatility, or unusually high customer-specific pricing may concentrate churn risk.

Potential action: proactive bill explanation, tariff comparison, or selective retention offer.

Business rationale: address price-related churn selectively rather than discounting broadly.

2 Renewal + relationship-depth play

Signal to validate: shorter-tenure customers appear more vulnerable; test whether renewal timing and product depth further concentrate risk.

Potential action: loyalty, contract, or bundle intervention for validated high-risk segments.

Business rationale: strengthen customer relationships where switching risk is genuinely elevated.

3 Behavioral early-warning queue

Signal to validate: combine churn risk with consumption, forecast deviation, and customer economics.

Potential action: prioritize outreach using both predicted churn probability and customer value.

Business rationale: focus retention spending where both churn risk and economic value are high.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features

Important caveats ensure the findings are interpreted as business signals, not causal proof

The analysis is strong enough to prioritize pilots, but not yet sufficient for final policy decisions.

Class imbalance

Only 9.72% of customers churned, so accuracy overstates practical performance.

Correlation vs causation

Relationships indicate association; they do not prove that a variable caused churn.

Missing churn timing

A single churn label limits month-by-month churn analysis and causal timing claims.

Low recall

The model caught 24 of 284 churners at the default threshold, limiting immediate operational usefulness.

Structural price zeros

Zero values in some tariff components may mean not applicable, not a true price.

Business validation needed

Retention offers must be evaluated against cost, margin, and customer response.

Source: PowerCo customer and pricing datasets; Random Forest baseline on engineered features